Department of Mathematics and Statistics, School of Arts and Science, Qatar University.

Unit Name:	Mathematics for EE	Instructor:	Dr. Mohamed Mabrok
Section:	L00	Location:	Lecture Room
Exam :	**Final Exam: Fall 2025**	Time and Date:	
Student Name:			
Student ID:			

- No credit will be given to answers for non-requested questions.
- Only non-plotting calculators are permitted. The use of the cell phones is absolutely prohibited.
- Shut off your cell phone completely.
- Make sure that you are feeling comfortable before entering the class for the test.
- Full mark will be given for complete answers for all questions.
- Cheating is an academic violation according to Qatar University rules and regulations, and in some cases, it may result in final dismissal from the University. Students should not under any circumstances commit or participate in any cheating attempt or any act that violates student code of conduct.

Question Number	Student Marks	Maximum Marks
1		**8**
2		**10**
3		**10**
4		**12**
5		**10**
6		**10**
7		**10**
Total / (Max)		**70**

1. Solve the following differential equation:

$$ye^x\,dx + (x+y^2)e^x\,dy = 0$$

8 Marks

2. Solve the following differential equation:

$$xy'' + (1-x)y' - y = x^2 e^x, \quad y_1(x) = e^x$$

10 Marks

3. Compute the following:

$$A)\ \mathscr{L}^{-1}\left(\frac{3s+8}{s^2+2s+5}\right)$$

$$B)\ \mathscr{L}\left(f(t)\right), \quad \text{where} \quad f(t)=\begin{cases} t & \text{if } t<6 \\ -8+(t-6)^2 & \text{if } t\geq 6 \end{cases}$$

10 Marks

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4. Using Laplace transforms, solve the folloing IVP

$$y'' + 5y' + 6y = f(t), \quad y(0) = 0, y'(0) = 0$$

where

$$f(t) = \begin{cases} t & \text{if } 0 \leq t < 2 \\ 5t + 5 & \text{if } t \geq 2 \end{cases}$$

12 Marks

5. Use Laplace transforms, find the general formula for solving the initial value problem for any $f(t)$ using the convolution.

$$y'' + 3y' = f(t), \quad y(0) = 0, \quad y'(0) = 0$$

Then, use the obtained general form and

$$f(t) = t$$

to find the solution (*You have to compute the integration*). 10 Marks

6. Show how the equation

$$y'' + 2y' + 5y = 0$$

can be transformed to the system

$$X' = \begin{bmatrix} 0 & 1 \\ -5 & -2 \end{bmatrix} X$$

Then, solve the system using matrix method. 10 Marks

7. Find the power series in x for the general solution of

$$(1+x^2)y'' + 6xy' + 6y = 0$$

and show that it can be given as:

$$\boxed{y = a_0(1 - 3x^2 + 5x^4 - 7x^6 + \dots) + a_1(x - 2x^3 + 3x^5 - \dots)}$$

10 Marks

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Algebra

- $\sin\theta = \frac{1}{\csc\theta}$
- $\csc\theta = \frac{1}{\sin\theta}$
- $\cos\theta = \frac{1}{\sec\theta}$
- $\sec\theta = \frac{1}{\cos\theta}$
- $\tan\theta = \frac{1}{\cot\theta}$
- $\cot\theta = \frac{1}{\tan\theta}$
- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$
- $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$
- $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$
- $\cos^2(x) = 1 - \sin^2(x)$
- $\sin^2(x) = 1 - \cos^2(x)$

Differentiation

- $\frac{\mathrm{d}}{\mathrm{d}x} c = 0$
- $\frac{\mathrm{d}}{\mathrm{d}x} x^n = n x^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} (ax \pm b)^n = a\, n\, (ax \pm b)^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|x| = \frac{1}{x}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|ax \pm b| = \frac{a}{ax \pm b}$
- $\frac{\mathrm{d}}{\mathrm{d}x} e^{ax} = a\, e^{ax}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \sin(ax) = a\, \cos(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} \cos(ax) = -a\, \sin(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} f(x)\, g(x) = f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x} + g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x}$ *Product rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} \frac{f(x)}{g(x)} = \frac{g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x} - f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x}}{(g(x))^2}$ *Quotient rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = f'(g(x)) \cdot g'(x)$ *Chain rule.*

Integration

- $\int a\, dx = ax + c$
- $\int x^n\, dx = \frac{x^{n+1}}{n+1} + c \qquad n \neq -1$
- $\int f(x)^n f'(x)\, dx = \frac{f(x)^{n+1}}{n+1} + c \qquad n \neq -1$
- $\int \frac{a}{x}\, dx = a\, \ln|x| + c$
- $\int \frac{1}{ax+b}\, dx = \frac{1}{a} \ln|ax+b| + c$
- $\int e^{ax}\, dx = \frac{1}{a}\, e^{ax} + c$
- $\int \frac{f'(x)}{f(x)}\, dx = \ln|f(x)| + c$
- $\int \sin(ax)\, dx = \frac{-1}{a}\, \cos(ax) + c$
- $\int \cos(ax)\, dx = \frac{1}{a}\, \sin(ax) + c$
- $\int \sec^2(ax)\, dx = \frac{1}{a}\, \tan(ax) + c$
- $\int \csc^2(ax)\, dx = -\frac{1}{a}\, \cot(ax) + c$
- $\int \tan(ax)\, dx = -\frac{1}{a}\, \ln(\cos(ax)) + c$
- $\int \cot(ax)\, dx = \frac{1}{a}\, \ln(\sin(ax)) + c$
- $\int \sec(ax)\tan(ax)\, dx = \frac{1}{a}\, \sec(ax) + c$
- $\int \csc(ax)\cot(ax)\, dx = -\frac{1}{a}\, \csc(ax) + c$
- $\int \sec(ax) dx = \frac{1}{a}\, \ln(\sec(ax) + \tan(ax)) + c$
- $\int \csc(ax)\, dx = \frac{-1}{a}\, \ln(\csc(ax) - \cot(ax)) + c$
- $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C$

Table of Laplace Transforms

	$f(t)=\mathfrak{L}^{-1}\{F(s)\}$	$F(s)=\mathfrak{L}\{f(t)\}$		$f(t)=\mathfrak{L}^{-1}\{F(s)\}$	$F(s)=\mathfrak{L}\{f(t)\}$
1.	1	$\frac{1}{s}$	2.	$\mathbf{e}^{at}$	$\frac{1}{s-a}$
3.	$t^n,\quad n=1,2,3,\ldots$	$\frac{n!}{s^{n+1}}$	4.	$t^p,\ p>-1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5.	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$	6.	$t^{n-\frac{1}{2}},\quad n=1,2,3,\ldots$	$\frac{1\cdot 3\cdot 5\cdots(2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7.	$\sin(at)$	$\frac{a}{s^2+a^2}$	8.	$\cos(at)$	$\frac{s}{s^2+a^2}$
9.	$t\sin(at)$	$\frac{2as}{(s^2+a^2)^2}$	10.	$t\cos(at)$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
11.	$\sin(at)-at\cos(at)$	$\frac{2a^3}{(s^2+a^2)^2}$	12.	$\sin(at)+at\cos(at)$	$\frac{2as^2}{(s^2+a^2)^2}$
13.	$\cos(at)-at\sin(at)$	$\frac{s(s^2-a^2)}{(s^2+a^2)^2}$	14.	$\cos(at)+at\sin(at)$	$\frac{s(s^2+3a^2)}{(s^2+a^2)^2}$
15.	$\sin(at+b)$	$\frac{s\sin(b)+a\cos(b)}{s^2+a^2}$	16.	$\cos(at+b)$	$\frac{s\cos(b)-a\sin(b)}{s^2+a^2}$
17.	$\sinh(at)$	$\frac{a}{s^2-a^2}$	18.	$\cosh(at)$	$\frac{s}{s^2-a^2}$
19.	$\mathbf{e}^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	20.	$\mathbf{e}^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$
21.	$\mathbf{e}^{at}\sinh(bt)$	$\frac{b}{(s-a)^2-b^2}$	22.	$\mathbf{e}^{at}\cosh(bt)$	$\frac{s-a}{(s-a)^2-b^2}$
23.	$t^n\mathbf{e}^{at},\quad n=1,2,3,\ldots$	$\frac{n!}{(s-a)^{n+1}}$	24.	$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right)$
25.	$u_c(t)=u(t-c)$ Heaviside Function	$\frac{\mathbf{e}^{-cs}}{s}$	26.	$\delta(t-c)$ Dirac Delta Function	$\mathbf{e}^{-cs}$
27.	$u_c(t)f(t-c)$	$\mathbf{e}^{-cs}F(s)$	28.	$u_c(t)g(t)$	$\mathbf{e}^{-cs}\mathfrak{L}\{g(t+c)\}$
29.	$\mathbf{e}^{ct}f(t)$	$F(s-c)$	30.	$t^n f(t),\quad n=1,2,3,\ldots$	$(-1)^n F^{(n)}(s)$
31.	$\frac{1}{t}f(t)$	$\int_s^\infty F(u)\,du$	32.	$\int_0^t f(v)\,dv$	$\frac{F(s)}{s}$
33.	$\int_0^t f(t-\tau)g(\tau)\,d\tau$	$F(s)G(s)$	34.	$f(t+T)=f(t)$	$\frac{\int_0^T \mathbf{e}^{-st}f(t)\,dt}{1-\mathbf{e}^{-sT}}$
35.	$f'(t)$	$sF(s)-f(0)$	36.	$f''(t)$	$s^2F(s)-sf(0)-f'(0)$
37.	$f^{(n)}(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$			